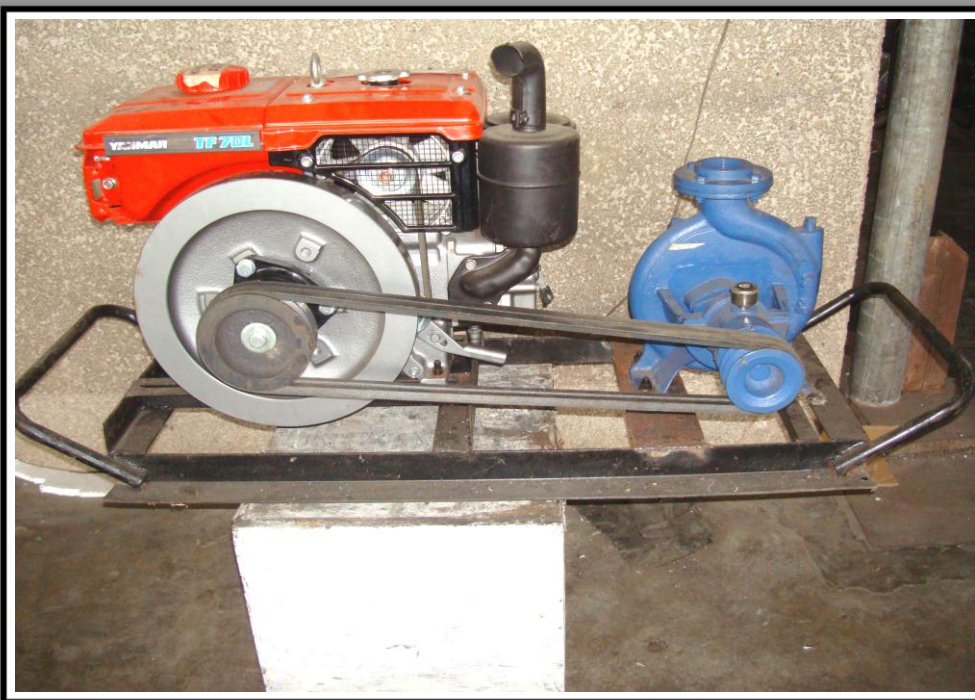




AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 99



**EAGLE EC50 Centrifugal Pump coupled to
YANMAR TF70L Diesel Engine**

EAGLE EC50 Centrifugal Pump coupled to YANMAR TF70L Diesel Engine - B

Test Report No.	:	2013 - 99
Date of Test	:	January 29, 2013
Test Valid Until	:	January 29, 2018
Brand	:	EAGLE
Model	:	EC50
Serial No.	:	No data
Type and Size	:	Single-end suction, 50 x 50 mm, Centrifugal pump
Manufacturer	:	SUPERCAST FOUNDRY & MACHINERY CORPORATION P. Gregorio St., Lingunan, Valenzuela City
Test Requested by	:	LAKAS AGRITECH MARKETING Rizal Street, Dagupan City Pangasinan
Machine Classification	:	Commercial

SUMMARY

The EAGLE EC50 centrifugal pump coupled to YANMAR TF70L diesel engine was tested for performance and suction condition. It was the second pumpset tested for the same brand and model. The first test was in 2012 with Test Report No. 2012 – 24. It was tested at the recommended full throttle speed setting of the engine. The engine and pump used double groove B-section pulleys with actual diameters of 89 mm (3.5 inches) and 76.2 mm (3.0 inches), respectively, and two pieces of V-belts. The engine and pump were mounted on a steel base.

At zero discharge condition, the Maximum Total Head (H) was 60.5 m. The shaft speeds of the pump and engine at this point were 3381 and 2357 rpm, respectively, with Engine Fuel Consumption Rate of 2.22 L/h. The Maximum Discharge Capacity was 11.3 L/s occurred at 6.75 m Total Head at shaft speeds of 2024 rpm and 1417 rpm for the pump and engine, respectively. The Fuel Consumption Rate at this point was 2.25 L/h.

The Maximum Fuel Consumption Rate of 2.30 L/h occurred at 6.84 L/s Discharge Capacity and at 29.6 m Total Head.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 7.74 m. The Net Positive Suction Head Available at this point was 2.13 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pumpset performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. **Figure 1** shows the pump mounted on the laboratory test set-up.

The test was conducted on January 29, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. EAGLE EC50 Centrifugal pump coupled to YANMAR TF70L Diesel engine on the test set-up

METHOD OF TEST

The pump was tested following the “ Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

Figure 2 shows the EAGLE EC50 single-end suction, 50 x 50 mm Centrifugal pump coupled to a 5.22 kW YANMAR TF70L Diesel Engine on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the engine pulley was 89 mm (3.5”), while that of the pump pulley was 76.2 mm (3.0”). Designed for a 50 mm diameter pipe, the suction inlet and discharge outlet were threaded flanges. The impeller was a closed type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing. Located on top of the pump beside the discharge outlet was the priming inlet. Located on top of the pump facing upward was the discharge outlet. The stuffing box used a packing seal. The pump shaft rotated on ball bearings kept in place by a bearing housing integral to the casing.

YANMAR TF70L Diesel engine, the primemover, was a four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized an oil bath type air cleaner. Lubrication system was force-feed lubricating type. Cooling was done through the circulation of water from the cylinder jacket to the condenser. It was started by a hand crank system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.

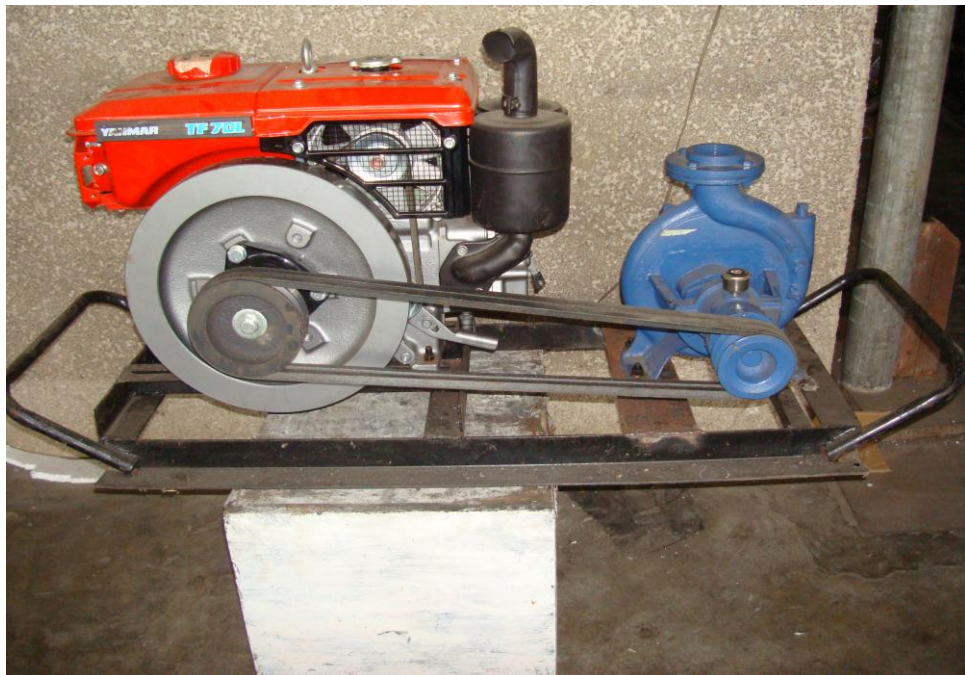


Figure 2. EAGLE EC50 Centrifugal Pump coupled to a YANMAR TF70L Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the EAGLE EC50 Centrifugal Pump coupled to a YANMAR TF70L Diesel Engine

1.	Brand	:	EAGLE
2.	Model	:	EC50
3.	Serial No.	:	No data
4.	Make	:	SUPERCASE FOUNDRY AND MACHINERY CORPORATIONS Philippines
5.	Type	:	Single-end suction, centrifugal pump
6.	Size (mm)	:	50 x 50
7.	Overall Dimension and Weight (Including mounting frame and engine)		
7.1	Length (mm)	:	1165
7.2	Width (mm)	:	499
7.3	Height (mm)	:	644
7.4	Weight (kg)	:	119.0
8.	Impeller		
8.1	Type	:	Closed
8.2	Diameter (mm)	:	187
8.3	Width (mm)	:	12
8.4	No. of vanes	:	8
8.5	Material	:	Brass
9.	Suction Inlet		
9.1	Type	:	Threaded flange
9.2	Diameter (mm)	:	50
9.3	Material	:	Cast Iron

Table 1. (Cont'n)

10.	Discharge Outlet		
10.1	Type	:	Threaded flange
10.2	Diameter (mm)	:	50
10.3	Material	:	Cast Iron
11.	Casing		
11.1	Type	:	Volute
11.2	Material	:	Cast Iron
12	Stuffing box	:	Packing seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	No data
15.	Rated Maximum Discharge Capacity (L/s)	:	No data
16.	Rated Maximum Total Head (m)	:	No data
17.	Primemover		
17.1	Brand	:	YANMAR
17.2	Model	:	TF70L
17.3	Serial No.	:	DC2114
17.4	Make	:	YANMAR CO., LTD., Indonesia
17.5	Type	:	Four stroke, water-cooled single horizontal cylinder diesel engine
17.6	Rated Output Power (kW)		
	Maximum	:	5.22 @ 2400 rpm
	Continuous	:	4.48 @ 2400 rpm

Table 1. (Cont'n)

17.8	Bore x Stroke	:	78 x 80
17.9	Displacement Volume (cc)	:	382
17.10	Cooling System	:	Water-cooled
17.11	Lubricating System	:	Force-feed type
17.12	Starting System	:	Manual cranking lever
17.13	Air Cleaner	:	Oil bath
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pump Tested : EAGLE EC50 coupled to
YANMAR TF70L diesel
engine

Date of Test : January 29, 2013

Test Condition

Ambient Temperature (°C) : 27.1
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (°C) : 27.4

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP- TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
2357	3381	2.22	60.5	0	0	21.6	0
2243	3212	2.24	54.4	1.61	0.86	21.7	3.95
2180	3023	2.21	50.7	2.56	1.27	21.4	5.92
2109	3019	2.24	46.5	3.53	1.61	21.7	7.43
2060	2936	2.27	43.0	4.20	1.77	22.0	8.04
1908	2727	2.28	36.4	5.47	1.95	22.2	8.82
1759	2525	2.30	29.6	6.84	1.99	22.3	8.92
1625	2323	2.28	22.8	8.31	.86	22.2	8.40
1504	2156	2.28	16.3	9.80	.57	22.1	7.10
1417	2024	2.25	6.75	11.3	0.75	21.8	3.43

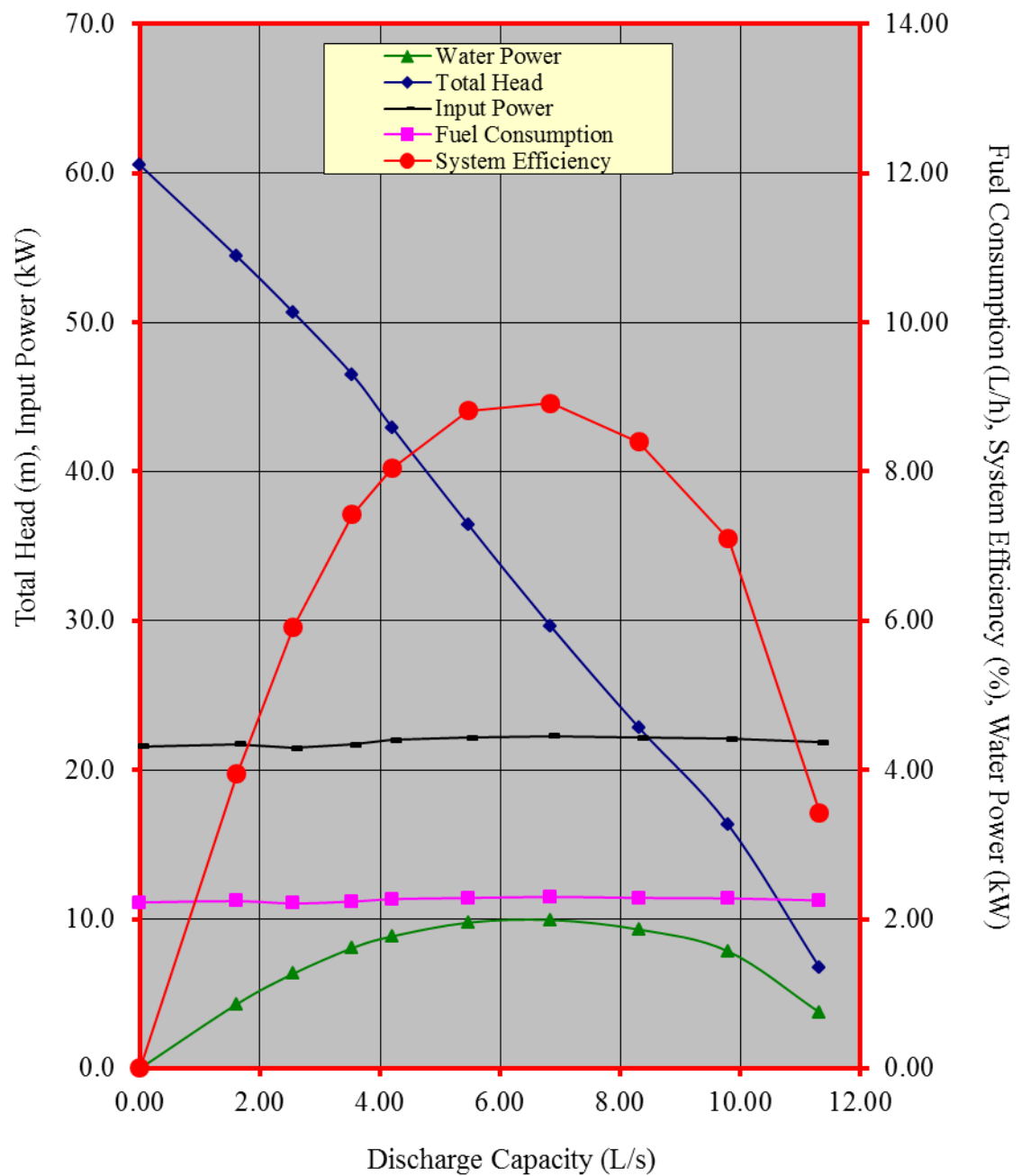


Figure 3. Performance characteristic curves of the EAGLE EC50 coupled to YANMAR TF70L Diesel Engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pump Tested	:	EAGLE EC50 Pump coupled to YANMAR TF70L diesel engine
Date of Test	:	January 29, 2013
Test Condition		
Ambient Temperature (°C)	:	28.3
Atmospheric Pressure (mb)	:	1013
Pumping Water Temperature (°C)	:	28.0

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
2029	1424	4.05	5.82	12.8	11.6
2588	1820	5.78	4.09	12.0	9.07
2845	1994	6.90	2.97	11.0	6.81
3655	2534	7.74	2.13	10.5	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

$$\text{Where: } H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
per second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power, WP , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power , IP , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determine by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

Test Engineer:

ROMULO E. EUSEBIO

Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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